

PADHAISHURU

GATE Preparation Platform

GATE CS

Computer Science & Engineering

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 34 questions (52%)

Difficult: 11 questions (17%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE CS — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
-

4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
-

Q1.

3M [-0.66]

[General Aptitude]

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative. Library 1st Main Road Canteen Physics Lab Hospital 5th Cross Road Chemistry Hostels Classrooms Lab N W E S

- (A) The library is located to the northwest of the canteen.
- (B) The hospital is located to the east of the chemistry lab.
- (C) The chemistry lab is to the southeast of physics lab.
- (D) The classrooms and canteen are next to each other.

Q2.

3M [-0.66]

[General Aptitude]

In the sequence 6, 9, 14, x, 30, 41, a possible value of is

- (A) 25
- (B) 21
- (C) 18
- (D) 20

Q3.

3M [-0.66]

[General Aptitude]

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and 2?

- (A) $1 + 2$ is not a prime number.
- (B) 12 is not a prime number.
- (C) $1 + 2 + 1$ is a prime number.
- (D) $12 + 1$ is a prime number.

Q4.

2M [no -ve]

[General Aptitude]

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500. The number of students who like their core branches is

Enter your answer here

Q5.

2M [-0.66]

[General Aptitude]

Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.

- (A) Kennel
- (B) Hammock
- (C) Hive
- (D) Lair

Q6.

2M [-0.66]

[General Aptitude]

If 'x' denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered _____ daunted]. Which one of the given options is appropriate to fill the blank?

- (A) phased
- (B) phrased
- (C) fazed
- (D) fused

Q7.

2M [no -ve]

[General Aptitude]

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

Enter your answer here

Q8.

2M [-0.66]

[General Aptitude]

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.

- (A) a; a; an
- (B) the; an; a
- (C) a; an; a
- (D) an; an; a

Q9.

2M [-0.66]

[General Aptitude]

Antonym of 'ephemeral':

- (A) Transient
- (B) Permanent
- (C) Brief
- (D) Short

Q10.

2M [-0.66]

[Engineering Mathematics]

Determinant of 2x2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q11.

1M [no -ve]

[Engineering Mathematics]

Probability of at least one head in 3 coin tosses:

Enter your answer here

Q12.

1M [-0.33]

*[Engineering Mathematics]*Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

- (A) 6
- (B) 7
- (C) 14
- (D) 15

Q13.

1M [-0.33]

*[Engineering Mathematics]*Derivative of x^3 at $x=2$:

- (A) 6
- (B) 12
- (C) 8
- (D) 4

Q14.

1M [-0.33]

*[Engineering Mathematics]*Mean of binomial $n=10, p=0.5$:

- (A) 2
- (B) 5
- (C) 10
- (D) 0.5

Q15.

1M [no -ve]

[Programming and Data Structures]

Infix to postfix uses:

Enter your answer here

Q16.

1M [-0.33]

[Programming and Data Structures]

What is the time complexity of inserting at the beginning of a linked list?

- (A) $O(n)$
- (B) $O(1)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Q17.

1M [no -ve]

[Programming and Data Structures]

Inorder traversal of BST gives:

Enter your answer here

Q18.

1M [-0.33]

[Programming and Data Structures]

Avg hash lookup:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q19.

1M [-0.33]

[Algorithms]

Kruskal uses:

- (A) DFS
- (B) BFS
- (C) Union-Find
- (D) Dijkstra

Q20.

1M [-0.33]

[Algorithms]

Merge sort space:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Q21.

1M [-0.33]

[Algorithms]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q22.

1M [no -ve]

[Algorithms]

BFS uses:

*Enter your answer here***Q23.**

2M [-0.66]

[Algorithms]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q24.

1M [-0.33]

[Algorithms]

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q25.

2M [no -ve]

[Operating Systems]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q26.

1M [no -ve]

[Operating Systems]

Semaphore with value 1 is:

Enter your answer here

Q27.

1M [-0.33]

[Operating Systems]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q28.

1M [-0.33]

[Operating Systems]

Round-robin uses:

- (A) Time quantum
- (B) Priority
- (C) Arrival time
- (D) Burst time

Q29.

1M [-0.33]

[Operating Systems]

Inode contains:

- (A) File name
- (B) Metadata
- (C) File data
- (D) Directories

Q30.

1M [no -ve]

[Operating Systems]

TLB stands for:

Enter your answer here

Q31.

1M [no -ve]

[Operating Systems]

Which of the following is NOT a necessary condition for deadlock?

Enter your answer here

Q32.

1M [-0.33]

[Computer Networks]

Protocol to find MAC from IP:

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) ICMP

Q33.

1M [-0.33]

[Computer Networks]

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP
- (B) UDP
- (C) HTTP
- (D) IP

Q34.

1M [no -ve]

[Computer Networks]

CSMA/CD is used in:

*Enter your answer here***Q35.**

1M [no -ve]

[Computer Networks]

Asymmetric encryption:

*Enter your answer here***Q36.**

1M [-0.33]

[Computer Networks]

OSPF is:

- (A) Distance vector
- (B) Link state
- (C) Path vector
- (D) Hybrid

Q37.

1M [no -ve]

[Computer Networks]

TCP connection establishment uses:

*Enter your answer here***Q38.**

1M [no -ve]

[Digital Logic]

Flip-flop stores:

Enter your answer here

Q39.

1M [-0.33]

*[Digital Logic]*The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q40.

1M [no -ve]

[Digital Logic]

Full adder has how many inputs:

Enter your answer here

Q41.

1M [-0.33]

[Theory of Computation]

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

Q42.

1M [no -ve]

[Theory of Computation]

PDA recognizes:

Enter your answer here

Q43.

1M [-0.33]

[Theory of Computation]

Which language is regular?

- (A) Equal 0s and 1s
- (B) Strings ending in ab
- (C) Palindromes
- (D) $a^n b^n$

Q44.

1M [-0.33]

[Compiler Design]

Lexical analyzer outputs:

- (A) Tokens
- (B) Parse tree
- (C) Intermediate code
- (D) Assembly

Q45.

2M [no -ve]

[CS]

The pseudocode of a function `fun()` is given below: `fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if (A[j]>A[j+1]) then swap A[j] and A[j+1] } Let [0, ... ,29] be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function fun() is called with [0, ... ,29] as argument, is _____. (Answer in integer)`

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Enter your answer here

Q46.

2M [no -ve]

[CS]

Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

Enter your answer here

Q47.

2M [no -ve]

[CS]

Consider the following grammar, with S as the start symbol. The grammar has three incomplete productions denoted by (1), (2), and (3). $S \rightarrow (1) \mid (2) (3)$ The set of terminals is $\{ , , , \}$. The FIRST and FOLLOW sets of the different non-terminals are as follows. $FIRST() = \{ , , \}$, $FIRST() = \{ , , \}$, $FIRST() = \{ , \}$ $FOLLOW() = \{ , , \$ \}$, $FOLLOW() = \{ \}$ Which one of the following options CORRECTLY fills in the incomplete productions?

- (A) (1) (2) (3)
- (B) (1) (2) (3)
- (C) (1) (2) (3)
- (D) (1) (2) (3)

Q48.

2M [-0.66]

[CS]

two semaphores: s_1 (initialized to 1) and s_2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below. // code of T1 // code of T2 $\text{wait}(s_1); \text{wait}(s_1); x = x+1; x = x+1; \text{print}(x); \text{print}(x); \text{wait}(s_2); \text{signal}(s_2); \text{signal}(s_1); \text{signal}(s_1);$ Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

- (A) T1 runs first and prints 1, T2 runs next and prints 2
- (B) T2 runs first and prints 1, T1 runs next and prints 2
- (C) T1 runs first and prints 1, T2 does not print anything (deadlock)
- (D) T2 runs first and prints 1, T1 does not print anything (deadlock)

Q49.

2M [-0.66]

[CS]

The value of the definite integral $\int_0^1 \int_0^1 \int_0^1 4x^2y - z^3 dz dy dx$ is . (Rounded off to the nearest integer)
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- (A) A
- (B) B
- (C) C
- (D) D

Q50.

2M [-0.66]

[CS]

Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____. GATE 2022 Computer Science and Information Technology (CS)

- (A) A
- (B) B
- (C) C
- (D) D

Q51.

2M [no -ve]

[CS]

Consider the following context-free grammar, where S , A , and B are the variables (non-terminals), and a , b , and c are the terminal symbols, S is the start variable, and the rules of G are described as: $S \rightarrow A | B | AB$ | Which ONE of the languages $L(G)$ is accepted by G ?

Enter your answer here

Q52.

2M [-0.66]

[CS]

The pseudocode of a function `fun()` is given below: `fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if (A[j]>A[j+1]) then swap A[j] and A[j+1] }` Let `[0, ..., 29]` be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function `fun()` is called with `[0, ..., 29]` as argument, is _____. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS1)

- (A) A
- (B) B
- (C) C
- (D) D

Q53.

2M [-0.66]

[CS]

$f: X \rightarrow Y$ is a function from X to Y , $g: Y \rightarrow Z$ is a function from Y to Z , and their composition defined as $(g \circ f)$ is a mapping from X to Z . If f and $g \circ f$ are onto (surjective) functions, which ONE of the following is TRUE about the function g ?

- (A) f must be an onto (surjective) function.
- (B) f must be a one-to-one (injective) function.
- (C) g must be a bijective function, that is, both one-to-one and onto.
- (D) g is not required to be a one-to-one or onto function.

Q54.

2M [-0.66]

[CS]

Which of the following statements about a relation R in first normal form (1NF) is/are TRUE?

- (A) R can have a multi-attribute key
- (B) R cannot have a foreign key
- (C) R cannot have a composite attribute
- (D) R cannot have more than one candidate key

Q55.

2M [-0.66]

[CS]

Consider a binary min-heap containing 105 distinct elements. Let i be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of i is

- (A) 53
- (B) 52
- (C) 27
- (D) 1

Q56.

2M [no -ve]

[CS]

Which ONE of the following techniques used in compiler code optimization uses live variable analysis?

Enter your answer here

Q57.

2M [-0.66]

[CS]

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

- (A) Only the root node
- (B) All leaf nodes
- (C) All internal nodes
- (D) Only the leftmost leaf node

Q58.

1M [no -ve]

[General Aptitude]

Synonym of 'ubiquitous':

Enter your answer here

Q59.

2M [-0.66]

[CS]

Let A be the adjacency matrix of the graph with vertices {1, 2, 3, 4, 5}. Let 1, 2, 3, 4, and 5 be the five eigenvalues of A. Note that these eigenvalues need not be distinct. The value of $1 + 2 + 3 + 4 + 5 =$.

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- (A) A
- (B) B
- (C) C
- (D) D

Q60.

2M [no -ve]

[CS]

Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Backpatching can be used to generate code for Boolean expression in one pass. (II) Backpatching can be used to generate code for flow-of-control statements in one pass. Which ONE of the following options is CORRECT?

Enter your answer here

Q61.

2M [-0.66]

[CS]

A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

- (A) E3
- (B) E1
- (C) E2
- (D) E5

Q62.

2M [-0.66]

[CS]

Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

- (A) $P = 4, Q = 10, R = 6, S = 2$
- (B) $P = 2, Q = 9, R = 5, S = 1$
- (C) $P = 4, Q = 12, R = 5, S = 4$
- (D) $P = 3, Q = 7, R = 7, S = 3$

Q63.

2M [no -ve]

[CS]

```
#include <stdio.h> int foo(int S[],int size){ if(size == 0) return 0; if(size == 1) return 1; if(S[0] != S[1]) return 1+foo(S+1,size-1); return foo(S+1,size-1); } int main(){ int A[]={0,1,2,2,2,0,0,1,1}; printf("%d",foo(A,9)); return 0; }
```

The value printed by the given C program is _____. (Answer in integer) Organising Institute: IIT Roorkee Computer Science and Information Technology (CS1)

- (A) A
- (B) B
- (C) C
- (D) D

Q64.

2M [-0.66]

[CS]

Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- (A) 10.12.2.0/23
- (B) 10.12.2.0/24
- (C) 10.12.0.0/22
- (D) 10.12.2.0/22

Q65.

2M [no -ve]

[CS]

Consider the following C function definition. `int f(int x, int y) { for (int i=0; i<y; i++) { x=x+x+y; } return x; }` Which of the following statements is/are TRUE about the above function?

Enter your answer here